Dhruv Walia

Department of Physics, Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram, Kerala, India
dhruvwalia.25@res.iist.ac.in | +91 7303515593 | ORCID: 0009-0000-0916-2338 | github.com/approxconst

RESEARCH INTERESTS

Condensed Matter Theory: Non-equilibrium quantum dynamics, Floquet systems, Eigenstate Thermalization Hypothesis (ETH) and ergodicity breaking, Topological phases, Driven quantum many-body systems.

EDUCATION

Indian Institute of Space Science and Technology (IIST) Thiruvananthapuram, India
Ph.D. in Theoretical Condensed Matter Physics (Advisor: Prof. Naveen Surendran) *Jan 2025 – Present*

- Coursework CGPA: 8.6/10.0

Nagoya University Nagoya, Japan
M.Sc. in Particle and Astrophysics (Advisor: Prof. Masayasu Harada) *Oct 2020 – Mar 2023*

- Thesis: *Construction of Neutron Star Matter in a SU(2) Parity Doublet Model with a Six-Point Interaction*

Nagoya University Nagoya, Japan
B.Sc. in Physics (Advisor: Prof. Chiho Nonaka) *Oct 2016 – Sep 2020*

- Research Project: *Signatures of Quark-Gluon Plasma in Relativistic Heavy-Ion Collisions* GPA: 3.47/4.30

RESEARCH PROJECTS

Doctoral Research – IIST Jan 2025 – Present

- **SSH model with nearest-neighbor interactions:** Using Exact Diagonalization to study the interacting Su-Schrieffer-Heeger chain. Quantum chaos appeared when interaction strength was close to the kinetic term.
- **Floquet engineering in dimerized chains:** Derived analytically the Floquet unitaries and effective stroboscopic Hamiltonians for step-drive (bang-bang) protocols in SSH model; found the phase diagram.
- **Quantum chaos in the extended Bose-Hubbard model:** Analyzed spectral statistics and tested ETH with next-nearest-neighbor interactions to reproduce well known results.

Project Student – North-Eastern Hill University (NEHU) Aug 2019 – Sep 2019
Department of Physics (Advisor: Prof. A. Saxena) *Shillong, India*

- Worked with Solid-State Nuclear Track Detectors (SSNTD): Performed chemical etching on materials after bombardment with heavy nuclei to characterize range and energy loss of heavy ions.

TEACHING EXPERIENCE

Teaching Assistant – IIST Jul 2025 – Present
Engineering Physics, Computational Physics; led lab sessions and problem-solving tutorials

Teaching Assistant – Nagoya University (during M.Sc.) Mar 2021 – Sep 2022
Quantum Mechanics, Statistical Physics, Analytical Mechanics, Mathematical Physics

HONORS, AWARDS & WORKSHOPS

- **Institute Research Fellowship**, IIST (Ph.D. grant) 2025 – Present
- **JASSO Scholarship**, Japan Student Services Organization 2016 – 2019
- **Rank 1, IESO Qualification Exam; Most Charismatic Participant, 9th IESO** (Brazil) 2015
- Winter School on Non-Local Correlations in Quantum Technology, TIFR Mumbai Feb 2026

TECHNICAL SKILLS

- **Methods:** Exact Diagonalization, Floquet theory, level-spacing statistics, ETH diagnostics, DMRG
- **Software & Programming:** Python, Julia, Fortran, C++, LaTeX; QuSpin, ITensor, NumPy, SciPy
- **Languages:** English (Fluent), Hindi (Native), Japanese (Conversational; JLPT N3)

LEADERSHIP & ACTIVITIES

Co-founder, HeForShe Student Chapter, Nagoya University
Active Member, Nagoya University Boxing Club